SBE37-SMP-ODO MicroCAT

Instrument Configuration

Instrument Serial Number:	37-16245
Instrument Firmware Version:	2.13.0
Zero Conductivity Frequency:	2828.70
Communications Format:	RS232
Communications Settings:	9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

Data Format	*Measurement*	*Sensor Type*	*Serial Number*	*Rating*
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10746412	2000m(2000 dBar)
RS232	Oxygen	SBE 63	63-1755	7000m

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.

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SENSOR SERIAL NUMBER: 16245
CALIBRATION DATE: 03-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

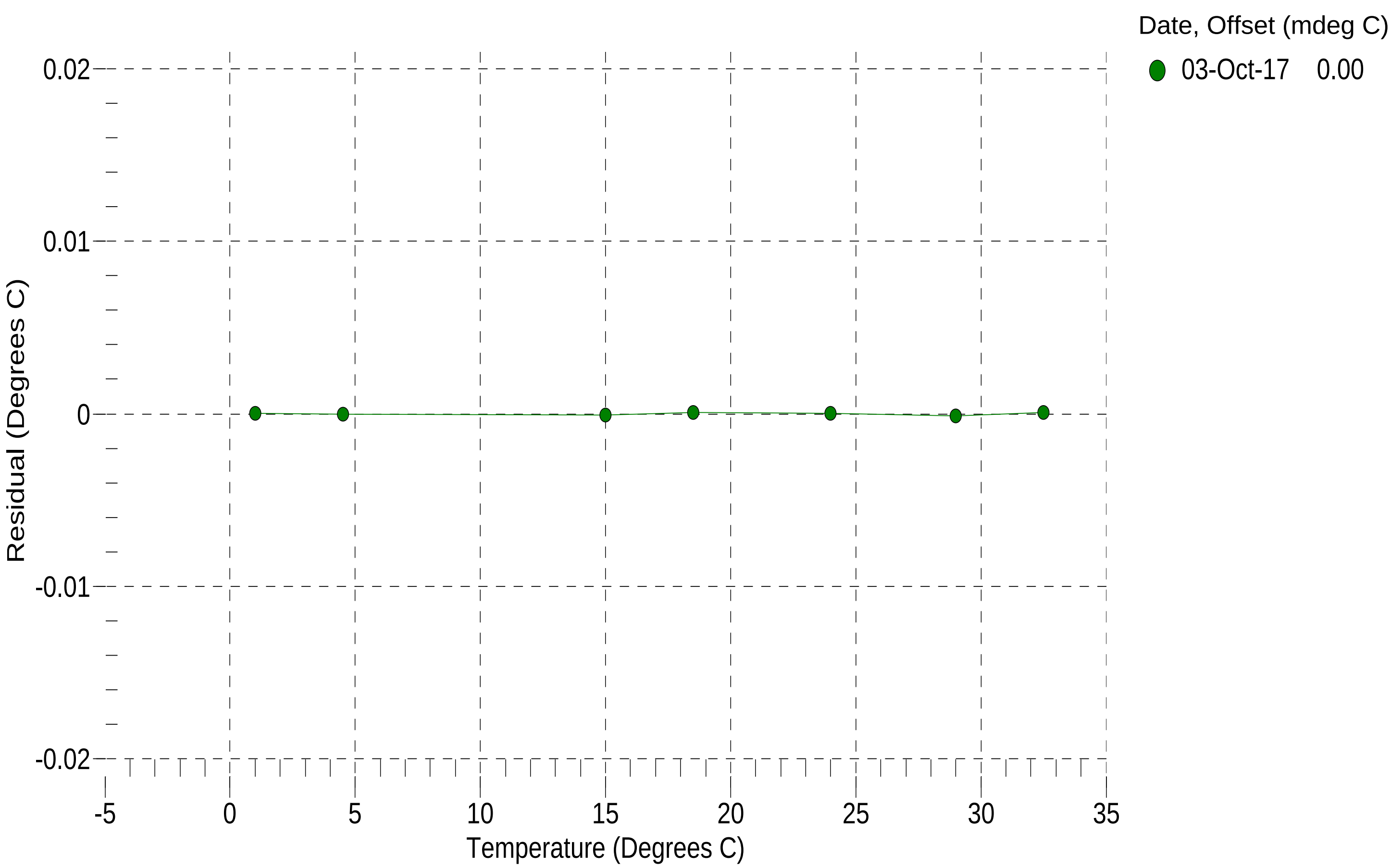
COEFFICIENTS:
a0 = -1.354808e-004
a1 = 3.119069e-004
a2 = -4.748925e-006
a3 = 2.074166e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	571142.0	1.0000	0.0000
4.5000	488658.6	4.5000	-0.0000
15.0000	312126.2	14.9999	-0.0001
18.5000	270489.6	18.5001	0.0001
24.0000	217293.2	24.0000	0.0000
29.0000	179162.8	28.9999	-0.0001
32.5000	157049.5	32.5001	0.0001

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1/\{a0 + a1[\mathit{ln}(n)] + a2[\mathit{ln}^2(n)] + a3[\mathit{ln}^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature



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CALIBRATION DATE: 03-Oct-17

SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -1.020533e+000
h = 1.276252e-001
i = -9.153850e-005
j = 2.192509e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = 2.9457e-007

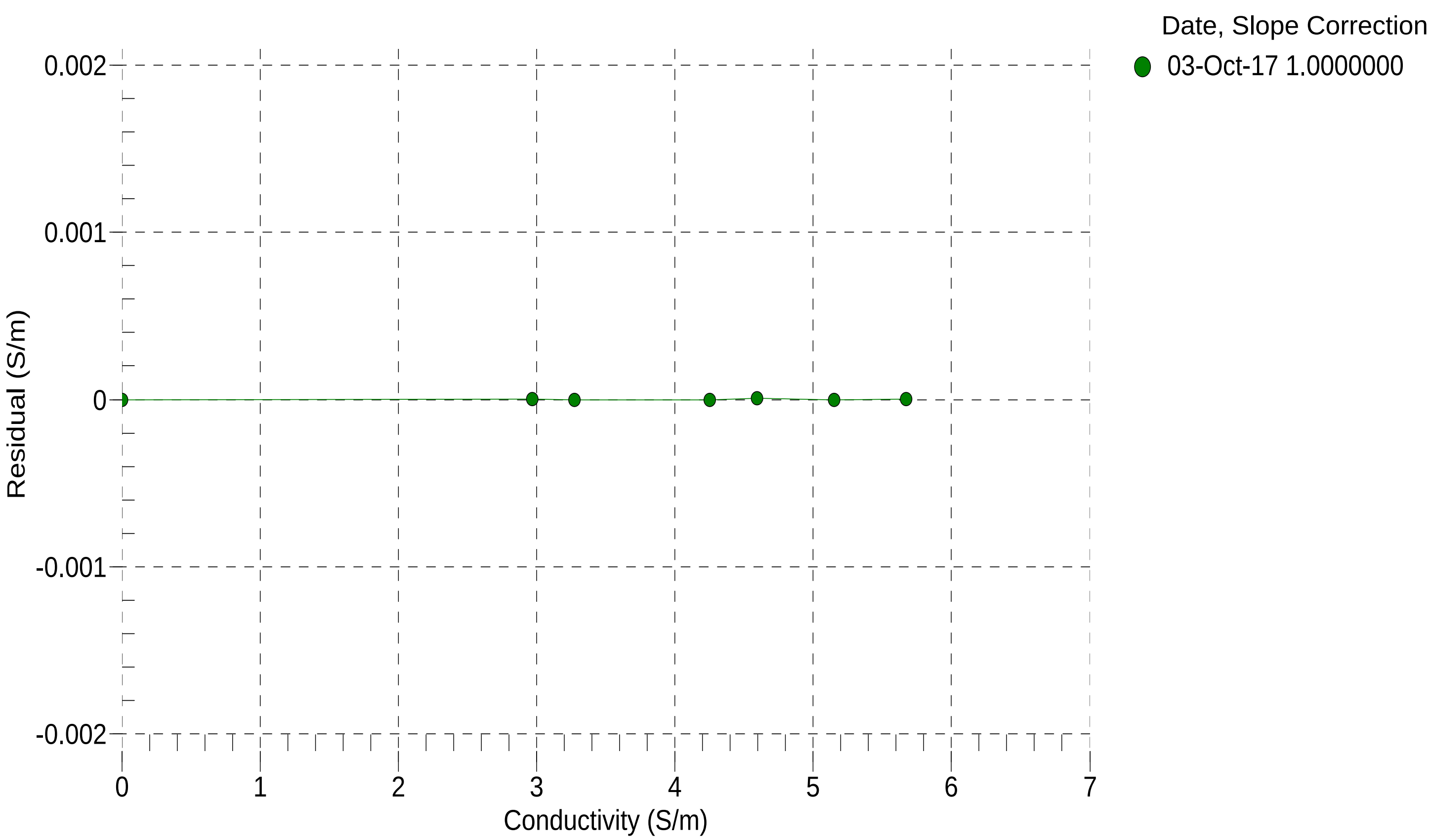
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2828.70	0.00000	0.00000
1.0000	34.6932	2.96644	5585.48	2.96645	0.00000
4.5000	34.6732	3.27255	5795.20	3.27255	-0.00000
15.0000	34.6306	4.25124	6419.20	4.25123	-0.00000
18.5000	34.6218	4.59534	6624.39	4.59535	0.00001
24.0000	34.6123	5.15162	6943.00	5.15162	-0.00000
29.0000	34.6073	5.67193	7228.01	5.67193	0.00000
32.5000	34.6050	6.04330	7423.47	6.04120	-0.00210

f = Instrument Output(Hz) * sqrt(1.0 + WBOTC * t) / 1000.0

t = temperature (°C); p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 16245
CALIBRATION DATE: 29-Sep-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10746412

COEFFICIENTS:

```
PA0 =       1.988571e-001
PA1 =       9.019128e-003
PA2 =      -1.503224e-010
PTEMPA0 = -5.904340e+001
PTEMPA1 =  5.427497e-002
PTEMPA2 = -5.943720e-007

PTCA0 =  5.232616e+005
PTCA1 = -6.146881e-002
PTCA2 =  4.465222e-002
PTCB0 =  2.507013e+001
PTCB1 = -1.745636e-004
PTCB2 =  0.000000e+000
```

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.59	524877.2	1530.6	14.58	-0.00
591.72	588935.6	1534.6	591.77	0.00
1168.84	653112.3	1535.3	1168.79	-0.00
1746.00	717444.9	1536.0	1745.97	-0.00
2323.13	781910.2	1536.9	2323.10	-0.00
2899.55	846439.3	1537.2	2899.54	-0.00
2323.07	781915.0	1536.9	2323.14	0.00
1746.03	717447.7	1536.4	1746.00	-0.00
1168.85	653119.5	1536.0	1168.86	0.00
591.68	588930.3	1535.6	591.72	0.00
14.59	524875.8	1536.1	14.56	-0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1719	524983.90
29.00	1652	524979.79
24.00	1557	524969.37
18.50	1452	524955.44
15.00	1385	524947.22
4.50	1186	524942.76
1.00	1120	524942.14

TEMPERATURE (°C)	SPAN (mV)
-5.00	25.07
35.10	25.06

y = thermistor output (counts)

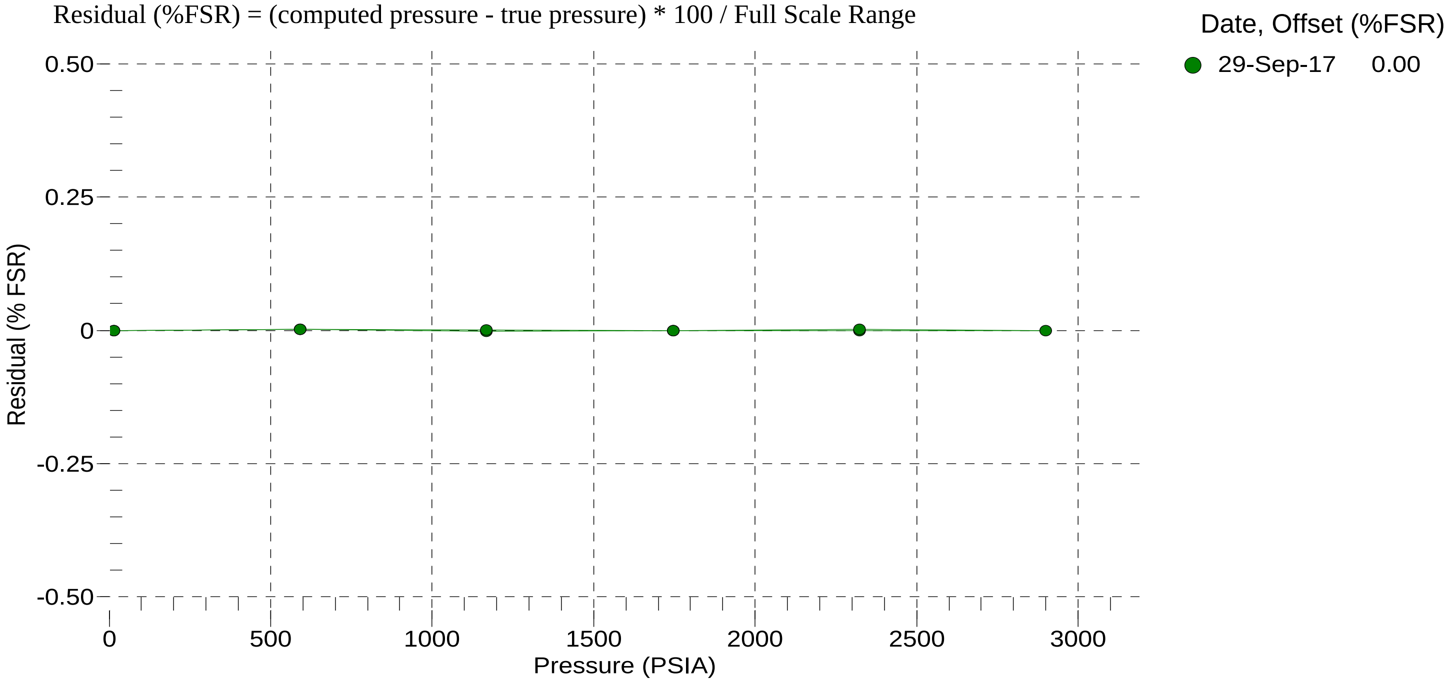
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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Pressure Test Certificate

Test Date: 2017-10-09

Description: SBE-37 Microcat

Sensor Information:

Model Number: SBE-37

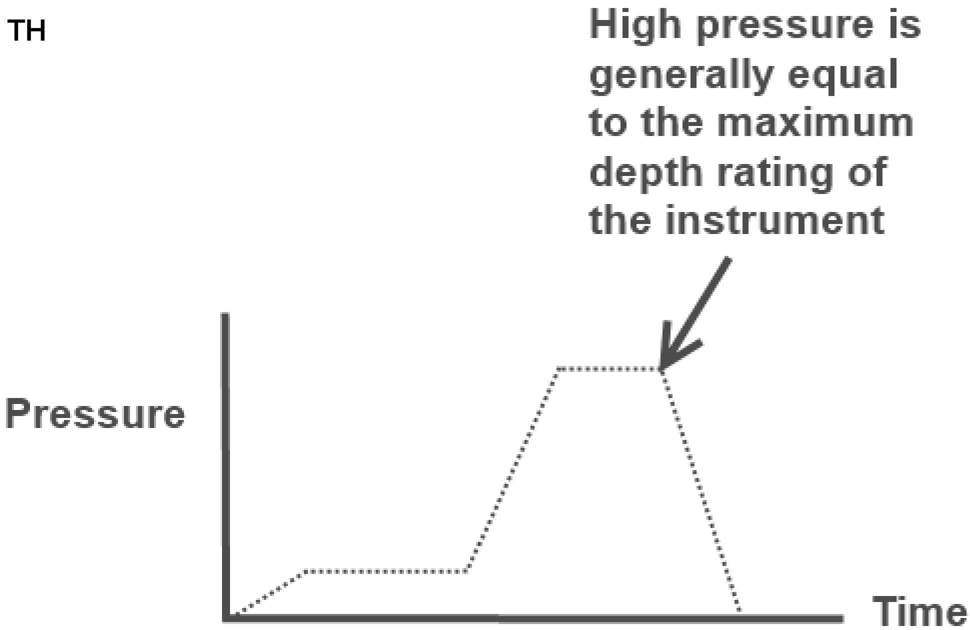
Serial Number: 16245

Pressure Test Protocol:

Low Pressure Test:	40	PSI	Held For: 15	Minutes
High Pressure Test:	2900	PSI	Held For: 15	Minutes

Passed Test: True

Tested By: TH



Typical Test Profile

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SENSOR SERIAL NUMBER: 1755
CALIBRATION DATE: 20-Sep-17

SBE 63 OXYGEN TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

TA0 = 7.266010e-004   TA2 = 1.381915e-006
TA1 = 2.438914e-004   TA3 = 7.859111e-008

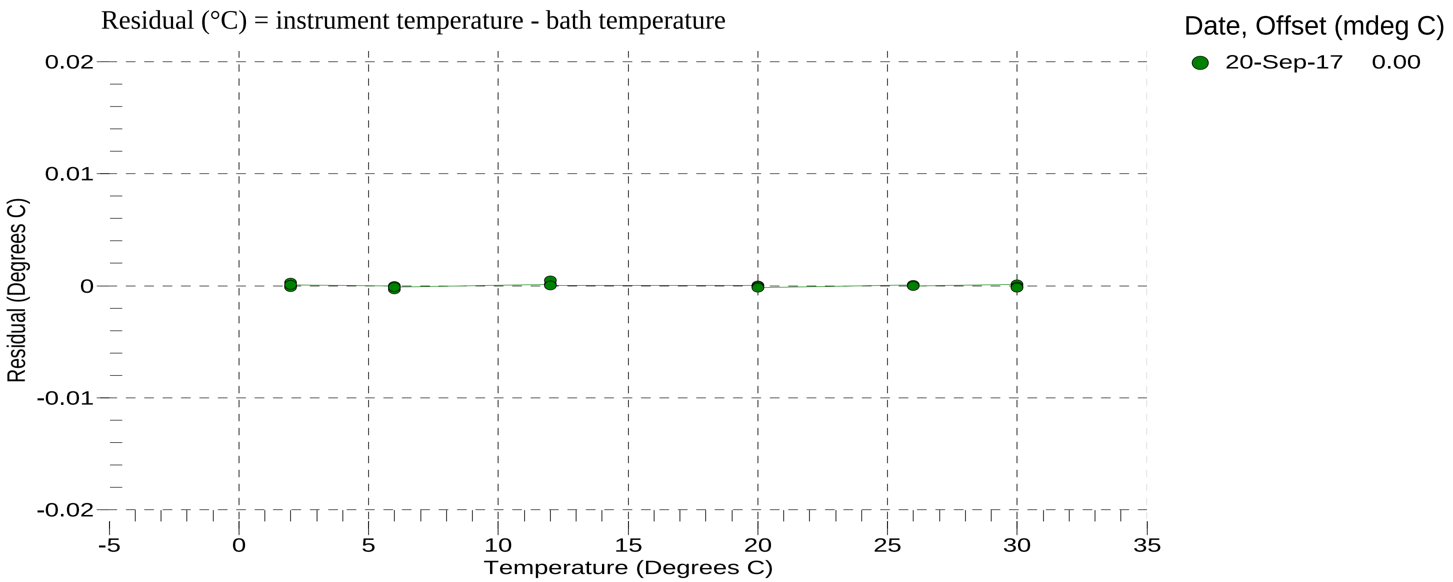
BATH TEMP (° C)	INSTRUMENT OUTPUT(V)	INST TEMP (° C)	RESIDUAL (° C)
1.9999	1.11858	1.9999	-0.00003
1.9999	1.11857	2.0002	0.00028
2.0000	1.11858	1.9999	-0.00013
2.0001	1.11857	2.0002	0.00008
6.0000	0.99438	6.0000	-0.00001
6.0000	0.99439	5.9997	-0.00035
6.0001	0.99438	6.0000	-0.00011
6.0001	0.99438	6.0000	-0.00011
12.0000	0.82856	12.0001	0.00010
12.0000	0.82855	12.0005	0.00049
12.0001	0.82856	12.0001	0.00000
12.0001	0.82856	12.0001	0.00000
19.9999	0.64496	19.9999	0.00004
20.0000	0.64496	19.9999	-0.00006
20.0001	0.64496	19.9999	-0.00016
20.0001	0.64496	19.9999	-0.00016
25.9999	0.53301	26.0000	0.00006
25.9999	0.53301	26.0000	0.00006
26.0000	0.53301	26.0000	-0.00004
26.0000	0.53301	26.0000	-0.00004
30.0000	0.46922	30.0001	0.00013
30.0000	0.46922	30.0001	0.00013
30.0002	0.46922	30.0001	-0.00007
30.0003	0.46922	30.0001	-0.00017

V = Instrument Output (Volts)

L = *ln* (100000 * V / (3.3 - V))

Temperature ITS-90 (°C) = 1 / (TA0 + (TA1 * L) + (TA2 * L^2) + (TA3 * L^3)) - 273.15

Residual (°C) = instrument temperature - bath temperature



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CALIBRATION DATE: 20-Sep-17

SBE 63 OXYGEN CALIBRATION DATA

COEFFICIENTS:

A0 = 1.0513e+000	B0 = -2.3936e-001	C0 = 9.7936e-002	E = 1.1000e-002
A1 = -1.5000e-003	B1 = 1.6646e+000	C1 = 4.1679e-003	
A2 = 4.4040e-001		C2 = 5.6324e-005	

BATH OXYGEN (ml/l)	BATH TEMPERATURE (° C)	BATH SALINITY (PSU)	INSTRUMENT OUTPUT (µsec)	INSTRUMENT OXYGEN (ml/l)	RESIDUAL (ml/l)
0.714	30.00	0.00	31.07	0.709	-0.005
0.732	26.00	0.00	31.80	0.728	-0.004
0.788	20.00	0.00	32.72	0.787	-0.001
0.874	12.00	0.00	34.09	0.872	-0.002
0.980	6.00	0.00	34.99	0.982	0.002
1.077	2.00	0.00	35.54	1.085	0.008
2.190	30.00	0.00	22.76	2.190	-0.000
2.325	26.00	0.00	23.29	2.326	0.000
2.468	20.00	0.00	24.44	2.471	0.003
2.982	12.00	0.00	25.33	2.978	-0.004
3.396	6.00	0.00	26.28	3.389	-0.007
3.643	30.00	0.00	18.80	3.645	0.003
3.749	2.00	0.00	26.92	3.741	-0.007
3.899	26.00	0.00	19.22	3.905	0.006
4.319	20.00	0.00	19.95	4.332	0.012
5.071	12.00	0.00	21.00	5.072	0.000
5.194	30.00	0.00	16.27	5.183	-0.012
5.607	26.00	0.00	16.58	5.605	-0.002
5.815	6.00	0.00	21.83	5.814	-0.001
6.221	20.00	0.00	17.22	6.228	0.006
6.432	2.00	0.00	22.41	6.438	0.006
7.286	12.00	0.00	18.17	7.283	-0.003
8.344	6.00	0.00	18.95	8.339	-0.005
8.858	2.00	0.00	19.81	8.861	0.003

T = temperature (°C) , P = pressure (dbar), U = Instrument output (µsec)

S_{corr} (salinity correction function) = 1.0 for calibration in DI water

See the user manual for more information on S_{corr} calculation

V = U / 39.457071

Oxygen (ml/l) = $\{((A0 + A1*T + A2 * V^2)/(B0 + B1 * V) - 1.0)/(C0 + C1 * T + C2 * T^2)\} * S_{corr} * \exp(E * P / (T + 273.15))$

Residual (ml/l) = instrument oxygen - bath oxygen